

MASTER OF COMPUTER APPLICATIONS (TWO YEAR) SEMESTER II EXAMINATION 2024-25
&
M.Sc. in COMPUTER SCIENCE SEMESTER II EXAMINATION 2024-25
CS - 322 : Deep Learning

Time : Three hours

Max. Marks : 70

(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)

Note: Question 1 is compulsory, attempt any four questions from the remaining six questions.

1. (a) The network consists entirely of CONV layers using 5×5 filters with stride 1 and *same* padding. The POOL layers use 3×3 max pooling with stride 2 (no padding). The number of filters and neurons are shown in brackets.

Input Image [128 x 128 x 3]	Conv1(32)	Conv2(64)	Pool_1	Conv3(64)	Conv4(32)	Pool_2	FC1(512)	FC2(128)	Output(5)
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- (I) The output size after pool_1. [1]
 (II) Number of parameters till pool_2 including bias. [2]
 (III) Total number of parameters from pool_2 till the output layer including bias. [2]
- (b) Which methods do we use to best fit the data in Logistic Regression? [1]
 (c) A 3-input neuron is trained to output a zero when the input is 110 and a one when the input is 111. After generalization, the output will be zero when and only when the input is? [2]
 (d) What are the vanishing and exploding gradients problems? [1]
 (e) After training a neural network, you observe a large gap between the training accuracy (100%) and the test accuracy (42%). Which of the method is commonly used to reduce this gap? [1]
 (f) You are building a neural network for spam vs. non-spam email classification. The final layer of your model has one output neuron producing a value z , which is first passed through a ReLU activation and then through a sigmoid function, giving $\hat{y} = \sigma(\text{ReLU}(z))$. You consider emails with $\hat{y} \geq 0.5$ as spam. What issue might this output function design cause during training or inference? [1]
 A. The sigmoid function will cancel out the ReLU and produce linear outputs.
 B. The output will never be exactly zero, making the model biased toward predicting spam.
 C. The gradient of the ReLU function is zero for negative z , which may block gradient flow during training.
 D. The model will not be able to classify any email as non-spam because sigmoid always outputs 1.
- (g) For a classifier with output logits $z=[2,1,0]$, compute the softmax probabilities. If the true class is the first class, compute the cross-entropy loss. [3]
2. (a) Explain the working procedure of a Recurrent Neural Network (RNN), including how it processes sequential data over time. [4]
 (b) Given an RNN with hidden state update $h_t = \tanh(Wx_t + Uh_{t-1} + b)$, where $W = [1]$, $U = [0.5]$, $b = 0$, $h_0 = 0$, and the input sequence is $x_1 = 1, x_2 = 2$, compute h_1 and h_2 . [4]
 (c) Given an LSTM with forget gate $f_t = \sigma(W_f x_t + U_f h_{t-1})$, cell state $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$, assume x_t is of size 4, W_f is of size 4×3 , U_f is of size 3×3 , $h_{t-1} = [0.5, 1.2, 2]$ and $C_{t-1} = [3, 4, 5]$. Randomly initialize W_f and U_f with values in the range $[0, 1]$. Compute f_t , i_t , $\tilde{C}_t$ (candidate cell state) and C_t . [6]
3. (a) Given an L-layer fully connected neural network with all weights $W^{[l]}=0$ and biases $b^{[l]}=0$, can the network learn meaningful representations during training using gradient descent? Justify your answer theoretically and with a concrete example. [4]
 (b) Discuss what is the impact of outliers on logistic regression? How to solve the problem of outliers. [4]
 (c) What issues can arise when using Mean Squared Error in place of Binary Cross-Entropy in logistic regression, particularly during training? [4]
 (d) What would happen if only linear activation functions were used in a multi-layer neural network, and why is this a limitation? [2]

P.T.O.

(2)

4. (a) A binary classification model was developed to detect whether patients have a certain disease (Positive) or not (Negative). The model was tested on a dataset of 1000 patients, and the following results were obtained: • True Positives (TP): 150 • True Negatives (TN): 800 • False Positives (FP): 30 • False Negatives (FN): 20
Compute the following performance metrics: (i) Accuracy (ii) Precision (iii) Recall (Sensitivity) (iv) F1-score (v) Interpret each metric you have computed. Explain what each metric indicates about the performance of the classifier in this context. (vi) If the cost of missing a positive case is very high (e.g., failing to detect a sick patient), which metric(s) should be prioritized in improving the model and why? [6]
- (b) Explain the concept of the Softmax Cross-Entropy loss function. How does it differ from Binary Cross-Entropy loss? Illustrate each with appropriate examples. [5]
- (c) Discuss how the choice of activation function affects gradient flow and model convergence in deep learning. Then, compute the output and derivative of the Leaky ReLU function ($\alpha = 0.01$) for an input value of -3. [3]
5. (a) Define transfer learning. How is it different from traditional machine learning approaches? [4]
- (b) What are the main components of a transfer learning system? Explain each with a brief example. [4]
- (c) Describe the concept of fine-tuning in transfer learning. How does it affect model performance? [3]
- (d) Explain the purpose and functioning of max pooling and average pooling in CNNs. [3]
6. (a) Explain the roles of the agent, environment, states, actions, rewards, and policy in reinforcement learning. [5]
- (b) Define a Markov Decision Process (MDP). [4]
- (c) What is the role of the reward function in reinforcement learning? [2]
- (d) What is a policy gradient method? Briefly describe how it works. [3]
7. Consider the following Convolutional Neural Network (CNN) architecture:
Input: $x^{(i)} \in \mathbb{R}^{[1 \times 256 \times 256]}$ (grayscale image)
Layer 1: $z_1 = \text{Conv}(x^{(i)}, W_1) + b_1$, $a_1 = \text{ReLU}(z_1)$, (Filter size: 3×3 , stride = 2, padding = 1, number of filters = 32)
Layer 2: $z_2 = \text{MaxPool}(a_1)$, (Pooling size: 2×2 , stride = 4)
Layer 3: $z_3 = \text{Conv}(z_2, W_2) + b_2$, $a_3 = \text{ReLU}(z_3)$, (Filter size: 4×4 , stride = 2, padding = 1, number of filters = 16)
Layer 4: $z_4 = \text{Flatten}(a_3)$, $a_4 = z_4$
Layer 5: $z_5 = W_3 a_4 + b_3$, $\hat{y}^{(i)} = \text{softmax}(z_5)$
Loss: $L^{(i)} = - \sum_{j=1}^C y_j^{(i)} \log(\hat{y}_j^{(i)})$, $J = \frac{1}{m} \sum_{i=1}^m L^{(i)}$ Based on the given information, answer the following questions:
- (a) What are the shapes of W_1 , b_1 , W_2 , b_2 , W_3 , and b_3 ? If we vectorize this network across m examples, what would the shape of each tensor be? [2]
- (b) How many trainable parameters does the model have in total? Show calculations per layer. [2]
- (c) If we add another convolutional layer after a_3 with 32 filters of size 3×3 and padding 1, how many additional parameters are introduced? [2]
- (d) Derive $\partial J / \partial \hat{y}^{(i)}$ and refer to it as $\delta_1^{(i)}$. Then write the expression for the full gradient $\partial J / \partial \hat{y}$. [2]
- (e) Derive $\partial \hat{y}^{(i)} / \partial z_5$ and refer to it as $\delta_2^{(i)}$. [1]
- (f) Derive $\partial z_5 / \partial W_3$ and $\partial z_5 / \partial a_4$. Refer to them as $\delta_3^{(i)}$ and $\delta_4^{(i)}$, respectively. [1]
- (g) Derive $\partial a_3 / \partial z_3$ and refer to it as $\delta_5^{(i)}$ (ReLU derivative). [1]
- (h) Derive $\partial z_3 / \partial W_2$ and refer to it as $\delta_6^{(i)}$. [1]
- (i) Derive the full gradient $\partial J / \partial W_2$ in terms of previous partial derivatives. Refer to it as $\delta_7^{(i)}$. [2]